

Assessing rumination response style among undergraduate nursing students: A construct validation study



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SUMMARY

Background: Rumination is one of the risk factors for mental health problems among nursing students. Rumination refers to repetitive thoughts focusing on negative emotions. There is a need to use an appropriate instrument to detect rumination to prevent the development of emotional problems. The ruminative response scale (RRS) is an instrument widely used to assess rumination levels in adult populations. It is inconclusive if the scale can be used for younger people like nursing students.

Objectives: This study aimed to examine the psychometric properties of the 10-item ruminative response scale (RRS-10) in undergraduate nursing students.

Design: A cross-sectional research design was used.

Settings: This study was undertaken at one of the universities in Thailand.

Participants: A sample of 747 undergraduate nursing students was recruited.

Method: Data were collected through a self-administered questionnaire, containing the 10-item RRS and Thoughts, Feelings and Experiences Questionnaire, and personal information. Exploratory/confirmatory factor analyses and correlational analyses were performed to test the construct and concurrent validity of the RRS respectively. Cronbach's alphas were used as parameters for the internal consistency reliability.

Results: The RRS-10 had two distinct factors: brooding (moody pondering) and reflection (analysing depression-relating situations). Adequate model fit indices were obtained and significant factor loadings were observed, indicating acceptable construct validity. These two factors had significant correlations with emotional distress, depression, anxiety, and hopelessness, supporting concurrent validity of the scale. Cronbach's alphas were 0.71 and 0.73 for the brooding and reflection factors respectively, suggesting good reliability.

Conclusion and recommendation: The RRS-10 demonstrated satisfactory psychometric properties. Therefore, it can be used to assess rumination response style among undergraduate nursing students in Thailand. Future research can test the scale on nursing students or youths in other countries.

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Introduction

Rumination is one of the cognitive vulnerability variables in the Response Style Theory of Depression. Rumination refers to repetitive and passive self-focused thoughts revolving around individuals' negative emotions and depression (Nolen-Hoeksema, 1991). Evidence indicates that rumination is a significant predictor of depression, anxiety, hopelessness, and psychological distress (Nolen-Hoeksema et al., 2008; Smith et al., 2006; Thanoi et al., 2010). In addition, two studies reported the relationship between rumination and suicidal ideation (Smith et al., 2006; Thanoi et al., 2010). To minimise the risks of the

emotional problems and suicidal ideation, it is essential to use an appropriate tool to early detect rumination.

Background

Undergraduate nursing students around the world are at risks of emotional problems (such as depression) (Klainin-Yobas et al., 2013; Prymachuk and Richards, 2008; Thanoi et al., 2012; Watson et al., 2008). Stress may contribute to such emotional problems given that nursing students generally encounter numerous stressors. Research has shown that younger nursing students encounter stress relating to their adolescence developmental stage (such as biological and hormonal changes) (Paperny, 2011) and academic stress (such as long study hours) (Burnard et al., 2008; Elcigil and Sari, 2007; Gorostidi et al., 2007; Hakojarvi et al., 2014; Prymachuk and Richards, 2007). Some students may ruminate over such stressful experiences, leading to emotional problems, depression, and suicidal ideation, the most serious

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consequence of rumination (Thanoi et al., 2010). Hence, there is a need to detect rumination levels among nursing students so that timely interventions can be provided to this population (Thanoi et al., 2012). The ruminative response scale (RRS) is the most widely used tool to assess individuals' rumination levels (Nolen-Hoeksema and Morrow, 1991).

The RRS was developed based on the Response Style Theory of Depression (Nolen-Hoeksema and Morrow, 1991). Within this theory, there are three main response styles: rumination, problem solving and distraction. People who are ruminating are fixated on their problems and associating negative feelings without taking actions. Rumination increases the risk for depression and other emotional problems via three mechanisms (Nolen-Hoeksema, 1991). Firstly, it enhances the recall of negative memories and pessimism. Secondly, it hinders an effective problem-solving process. Lastly, rumination inhibits instrumental behaviours and therefore reducing individual's sense of control (Nolen-Hoeksema, 1991).

The RRS was developed as part of the Response Style Questionnaire and the original RRS had 22 items (RRS-22) capturing three components: self-focused, symptom/depression-focused and cause/consequence-focused thoughts (Nolen-Hoeksema and Morrow, 1991). A major criticism of RRS is that items concerning symptom/depression-focused thoughts overlap with measures of clinical depression. This might explain a spuriously high correlation between the RRS-22 and measures of depression. Subsequently, Treynor et al. (2003) modified the RRS-22 by removing such overlapping items, resulting in the 10-item RRS (RRS-10). Following factor analyses, two emerging factors of the RRS-10 were labelled Reflection and Brooding (Treynor et al., 2003). The former suggests a purposeful thought to find ways to alleviate one's depressive symptoms whereas the latter indicates a passive thought to compare persons' current situation with some unrealistic standard (Treynor et al., 2003). The RRS-10 is perceived to be easier to use, less time-consuming to complete, less burden on participants, and more economical than the RRS-22.

A screening tool requires acceptable psychometric properties such as construct validity, concurrent validity, and reliability. Construct validity refers to the extent to which the scale captures the underlying construct it is supposed to measure (Streiner and Norman, 2008). Factor analyses can provide evidence to support the construct validity (Streiner and Norman, 2008). Concurrent validity is supported by correlation coefficients between the scale and other gold standard measures (Streiner and Norman, 2008). Psychometric properties of the RRS-22 have been established in English-speaking individuals (Luminet, 2004). Furthermore, the scale has been cross-validated in Japanese (Ito and Agari, 2002), Dutch (Raes et al., 2003), and Turkish samples (Baker and Bugay, 2012). Nevertheless, such cross-validation is limited for the RRS-10 and it is unclear if the scale will be useful for people in other countries like Asian adolescent samples (such as undergraduate students). This indicates a knowledge gap in the existing literature.

Study Aims

Consequently, this study aimed to examine psychometric properties (construct validity, concurrent validity, and internal consistency reliability) of the RRS-10 on Thai undergraduate nursing students. Two emerging research questions were: a) what is the factorial structure of the RRS-10? and b) Is the scale correlated with other standardized measurements?

Guided by the Response Style Theory of Depression and previous empirical evidence (Treynor et al., 2003), the first hypothesis addressed that the scale would display the two-factor structure: Reflection and Brooding and the two emerging factors would have acceptable reliability. Theoretically, rumination is postulated to have a positive association with various emotional problems. Therefore, the second hypothesis stated that the RRS-10 (the measure of rumination) would significantly

correlate with the measure of emotional distress (depression, anxiety, and hopelessness). Positive findings from this study might suggest that the RRS-10 is a valid and reliable tool to screen rumination levels among undergraduate nursing students.

Methods

Research Design

A cross-sectional research design was used to test the psychometric properties of the RRS-10. The research design enabled assessments of the scale validity, the conceptual structure, and reliability (Grove et al., 2013).

Target Population and Sample

The target population included nursing students who are enrolled in the four-year undergraduate programme (Nursing and Midwifery) at a university in Thailand. Both male and female nursing students were eligible regardless of their academic year. No exclusion criteria were set as the researchers aimed to recruit all students. During the recruitment period, there were approximately 1200 enrolled nursing students. Following the ethics approval, the investigators seek permission from the Dean to collect data on nursing students. Subsequently, the investigators scheduled out-of-class meetings with groups of nursing students to provide information about the study, allow time to answer questions they may have and request for their participation. Interested students were requested to sign a consent form, complete a self-reported questionnaire, and then submit it to the investigators.

Ethical Consideration

This study obtained the ethics approval from the Internal Review Board (IRB) of the university. According to the IRB, participants were required to sign a consent form before completing the questionnaire. The investigators emphasized the issues of voluntary participation (i.e., students were not obligated to taking part in the study and there would be no penalty for non-participants) and anonymity during the data collection. To protect the identity of the study participants, they were advised not to write their personal information on the questionnaire. Furthermore, the signed consent forms were kept separately from the completed questionnaires and there were no links between the two documents. Therefore, it was not possible to trace back to participants.

Variables and Measurements

Data were collected using anonymous self-reported questionnaires that take about 20 min to complete. The questionnaire contained information listed below.

Rumination was measured with the RRS-10 (Treynor et al., 2003). Participants were asked to rate on four response categories ranging from "never/almost never" (1) to "always/almost always" (4). Possible scores are in the range of 10–40 with higher scores signifying higher levels of ruminative response style. On a sample of elderly living in communities in the USA ($n = 1328$), factor analyses revealed that the scale had two main factors: Reflection and Brooding (Treynor et al., 2003). Cronbach's alphas were 0.60–0.72 for the Reflection factor and 0.62–0.77 for the Brooding factor (Treynor et al., 2003). The RRS-10 was translated into Thai language using Brislin's (1986) translation method. Particularly, a nurse educator specialized in mental health nursing translated the scale into Thai and another nurse educator back-translated the Thai version into English. Next, the two versions were examined for semantic and content equivalence by five native English speakers.

Emotional distress was measured with the 18-item Thought, Feeling, and Experience Questionnaire (TFEQ), devised to capture depression, hopelessness, and anxiety. The scale items were derived from the Measurement of Adolescent Potential for Suicide (MAPS; Eggert et al., 1994b) and High School Questionnaire (HSQ; Eggert et al., 1994a, 1994b). The TFEQ was translated by Thanoi et al. (2011) into Thai using Brislin's (1986) translation technique. The depression subscale has seven items (such as "I felt lonely"). The anxiety subscale consisted of four items that assessed physical anxiousness, recurrent frightening thoughts, jumbled thoughts and feelings of uneasiness. The hopelessness subscale comprises seven items (such as "I felt hopeless about my life"). The response formats range from "never" (0) to "always" (6) and total scores are in the range of 0–108. High scores reflect high levels of emotional distress. In Thai adolescent sample, Cronbach's alphas of the TFEQ was 0.90, suggesting excellent reliability (Thanoi et al., 2011).

Personal information was also collected using a series of close- and open-ended questions. This information encompassed age, gender, education, grade, parents' marital status, living arrangement, family income, and their monthly allowance.

Data Analyses

The statistical analyses were performed using SPSS 17.0 and IBM SPSS AMOS 21. Descriptive statistics (e.g., mean and standard deviation) were used to describe the sample and examine study variables. To test the construct validity of the RRS-10, exploratory factor analyses (EFA) were conducted, followed by confirmatory factor analyses (CFA) using AMOS. Model fits of the factorial model were determined by: Chi-square, root mean square error of approximation (RMSEA) and Comparative Fit Index (CFI) (Hair et al., 2006). Internal consistency reliability of the derived factorial model was determined by Cronbach's alpha. To assess concurrent validity, Pearson Product Moment Correlation analyses were undertaken: the resulting association between the RRS-10 and the measure of emotional distress was then examined.

Results

Approximately 1200 nursing students were enrolled in the Faculty of Nursing of the study university; however, only 830 expressed their interest to take part in this study. A total of 830 questionnaires were distributed to these nursing students but only 747 completed ones were returned, making up a response rate of 62.25% (747/1200). An average age of the students was 20.53 years (SD = 1.55). Most participants were female (96.50%; $n = 721$), were in Junior year (27.30%, $n = 204$), had grade point average from 2.51 to 3.00 (41.63%, $n = 311$), and had sufficient family income (86.30%, $n = 645$). The details are shown in Table 1.

Construct Validity of the RRS-10

Exploratory factor analyses of the RRS-10 were performed using principal axis factoring (PAF) and Varimax rotation method. Table 2 displays two underlying factors (Reflection and Brooding) and factor loadings of questionnaire items (represented by lambda or λ). Factor loadings denote the correlation between each questionnaire item and its respective underlying factor (Streiner and Norman, 2008). Factor loadings (λ) greater than 0.30 are considered sufficient and this could suggest that the item adequately captures the underlying factor (Streiner and Norman, 2008).

Results (Table 2) suggested that the RRS-10 had two distinct factors: Reflection and Brooding. Each factor contained five questionnaire items. On the Brooding factors, items 1, 3, 6, 7, and 8 had factor loadings (λ) more than 0.30 with the range of 0.46–0.81. Items 2, 4, 5, 9, and 10 had acceptable factor loadings on the Reflection factor and the factor loadings ranged from 0.59 to 0.76. Evidence

Table 1
Demographic characteristics of sample ($n = 747$).

Demographic characteristics	Values	Number	Percent
Gender	Male	26	3.50
	Female	721	96.50
Age (year)	18–19	191	25.60
	20–21	372	49.80
	>21	179	24.00
	Missing	5	0.60
	Freshman	163	21.80
Academic year	Sophomore	186	24.90
	Junior	204	27.30
	Senior	194	26.00
Grade point average	<2.00	5	0.67
	2.00–2.50	85	11.38
	2.51–3.00	311	41.63
	3.01–3.50	287	38.42
	>3.50	36	4.82
	Missing	23	3.08
Monthly allowance Thai Baht (US\$)	<5000 (<US\$ 160)	446	59.71
	5001–10,000 (US\$ 160–315)	290	38.82
Sufficiency of family income	More than 10,000 (>US\$ 315)	6	0.80
	Missing	5	0.67
	Sufficient	645	86.35
	Insufficient	101	13.52
	Missing	1	0.13

from exploratory factor analyses suggested that the RRS-10 contained items that adequately measured two components of rumination (Reflection or Brooding). This information supported the construct validity of the RRS-10.

Subsequently, the aforementioned two-factor model was submitted to confirmatory factor analysis using AMOS. The resulting significant chi-square ($\chi^2 = 191.132$; Df = 33; $P < 0.001$) indicated a bad fit between the model and sample data. However, it is well-known that chi-square is sensitive to sample size and any trivial mismatch can achieve statistical significance (Kline, 2011). Therefore, we examined other fit indices and found that they revealed acceptable fit; Comparative Fit Index = 0.91, RMSEA = 0.06, and 90% Confidence Interval of RMSEA 0.07–0.09. These fit indices suggested that the two-factor model of the RRS-10 was acceptable.

Table 2
Results from exploratory factor analyses ($n = 747$).

Questionnaire items	Factor loading (λ)	
	Reflection factor	Brooding factor
RM1. Think "What am I doing to deserve this?"		0.63
RM3. Think "Why do I always react this way?"		0.46
RM6. Think about a recent situation, wishing it had gone better		0.52
RM7. Think "Why do I have problems other people don't have?"		0.80
RM8. Think "Why can't I handle things better?"		0.81
RM2. Analyse recent events to try to understand why you are depressed	0.72	
RM4. Go away by yourself and think about why you feel this way	0.67	
RM5. Write down what you are thinking and analyse it	0.59	
RM9. Analyse your personality to try to understand why you are depressed	0.76	
RM10. Go someplace alone to think about your feelings	0.66	

Note:

a) Principal axis factoring and Varimax rotation methods were used.

b) Numbers in the table are factor loadings (λ), which represent the correlation between each questionnaire item and its respective underlying factor.

c) Items with factor loadings greater than 0.30 are considered acceptable measures of the underlying factors.

Table 3
Correlations among study variables (n = 747).

Variable	RRS-10 [Total]	Reflection factor	Brooding factor	Emotional distress	Depression subscale	Anxiety subscale
Reflection factor	0.90**					
Brooding factor	0.86**	0.54**				
Emotional distress [Total]	0.52**	0.39**	0.54**			
Depression subscale	0.54**	0.43**	0.52**	0.94**		
Anxiety subscale	0.49**	0.70**	0.50**	0.84**	0.76**	
Hopelessness subscale	0.32**	0.23**	0.42**	0.87**	0.71**	0.57**

** Significance at the 0.01 level.

Results in Fig. 1 confirmed that the RRS-10 had two distinct factors (Brooding and Reflection) and all questionnaire items were acceptable measures of their respective factors. All factor loadings are statistically significant at the significance level of 0.01. The Brooding factor had five items and factor loadings of the items ranging from 0.49 to 0.62. The reflection factor also had five items whose factor loadings were in the range of 0.43–0.71. In addition, the two factors were significantly correlated with each other ($r = 0.75$; $P < 0.01$). Evidence from confirmatory factor analyses further confirmed that the RRS-10 had items that capture two components of rumination (Reflection and Brooding).

As mentioned earlier, construct validity refers to the degree to which the scale captures the underlying construct it is supposed to measure (Streiner and Norman, 2008). The construct of “rumination” refers to repetitive and passive self-focus thoughts surrounding one’s depression and emotional problems (Nolen-Hoeksema and Morrow, 1991). Furthermore, the RRS-10 was devised to capture rumination with the proposition that the scale has two major components: Reflection and Brooding (Treyner et al., 2003). This study provided empirical evidence to support Treyner et al.’s proposition and, therefore, supported the construct validity of the RRS-10. Specifically, our findings showed that items in the RRS-10 captured the rumination construct and that the scale covered two components (factors) of rumination. Additionally, the Reflection component contained items that assess how individuals keep analysing depression-related situations. The Brooding component had items that measure how the persons continue thinking about their emotional problems.

Reliability

Reliability analyses were undertaken to assess internal consistency reliability of the RRS. Findings suggested that the overall RRS-10 had good reliability ($\alpha = 0.80$). Similarly, the Reflection ($\alpha = 0.73$) and Brooding subscales ($\alpha = 0.71$) also had acceptable reliability.

Concurrent Validity of the RRS-10

Correlation analyses were used to test the concurrent validity of the RRS-10. It appeared that the RRS-10 total score had significant positive correlations with the emotional distress (total score); and the depression, anxiety, and hopelessness subscale (Table 3). Similarly, both Reflection and Brooding factors had significant positive associations with emotional distress (total score) and all the subscales. Correlation coefficients for the Reflection factor ranged from 0.23 to 0.70 and those for the Brooding factor were in the range of 0.42–0.54. This evidence supported the concurrent validity of the RRS-10.

Discussion

This study aimed to examine the psychometric properties of the RRS-10 on nursing students in Thailand. A sample of 747 Thai

undergraduate nursing students was recruited. Results indicated that the RRS-10 has acceptable construct validity, concurrent validity, and reliability. Therefore, it can be used as a screening tool to assess rumination levels among the students.

The first study hypothesis was supported by results of this study. Particularly, the RRS-10 had two distinct factors: Reflection and Brooding. Items (with similar contents and meaning) were clustered together and appeared to capture the underlying factor reasonably well. This is an evidence to support the construct validity of the RRS-10. The Reflection factor contained such items as “analysing recent event to understand why feeling depressed”, “analysing own personality to try to understand why feeling depressed”, and “write down the thought and analyse it”. The Brooding factor encompassed the items like “what am I doing to deserve this”, “why can’t I handle things better” and “why do I have problem that people don’t have.” To our perspectives, people with reflection tend to overanalyse relevant situations, events, and personality; and try to understand the cause of their depression. Those with Brooding keep pondering why unfortunate events happen to them. Our findings supported Treyner et al.’s (2003) work concerning the two-factor structure of the RRS-10. In addition, the RRS-10 and the two factors had acceptable reliability and these findings agreed with previous studies (Baker and Bugay, 2010; Baker and Bugay, 2012; Treyner et al., 2003).

According to the Response Style Theory of Depression, people who are ruminating have the tendency to experience various emotional problems (Nolen-Hoeksema, 1991). Hence, the measurement of rumination should be positively correlated with those of negative emotions. Our findings supported the second study hypotheses and were in line with the theory and conceptualization. The RRS-10 and the two factors (Reflection and Brooding) showed significant positive associations with measures of emotional distress, depression, anxiety, and hopelessness. This evidence supported the concurrent validity of the RRS-10. Our findings were consistent with previous studies. Specifically, the RRS-10, Reflection factor and Brooding factor were significantly correlated with measures of stress, worry (Nikcevic et al., 2014), depression (Baker and Bugay, 2012; Treyner et al., 2003), and psychiatric symptomatology (such as anxiety, phobic anxiety, obsessive-compulsive disorder, somatization, and interpersonal sensitivity) (Baker and Bugay, 2010).

This study was strengthened by the use of CFA (with AMOS), which provided an elegant way to examine the construct validity of the RRS-10 using goodness-of-fit indices and enhance accuracy in estimating crucial parameters (i.e., factor loadings and correlation coefficients). Furthermore, the large sample size ($n = 747$) might increase the generalizability of research findings to nursing students in the university. Nevertheless, some methodological limitations were noted for this study. Specifically, the use of single-centre recruitment might limit the external validity to nursing students in other universities and to youth in other settings. Furthermore, gender comparisons could not be conducted given that there were only a few male students in a university. Additionally, this research

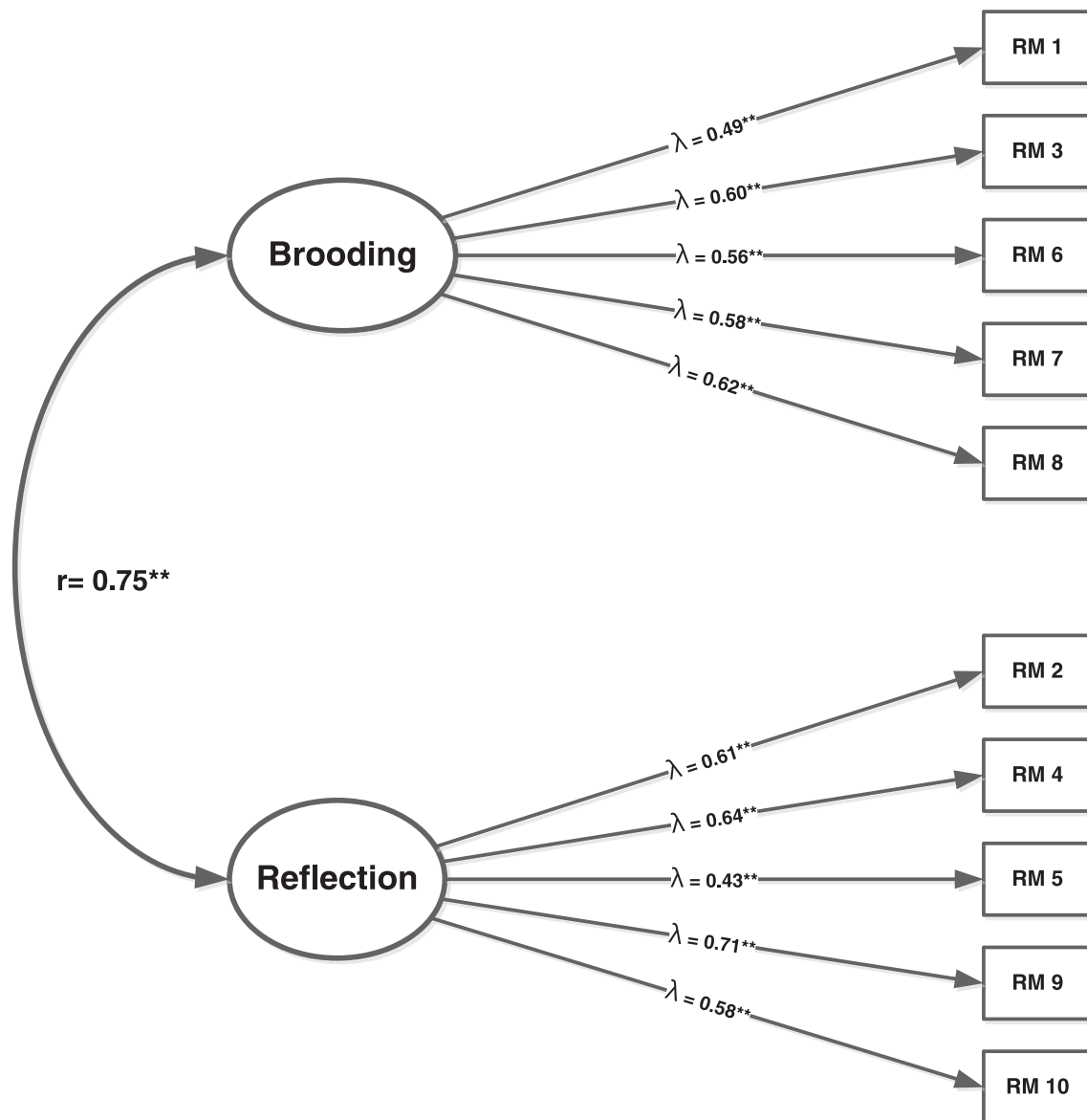


Fig. 1. Results from confirmatory factor analysis. Note: a) Factor loading (λ) denotes a correlation between each questionnaire item and its respective underlying factor. b) Items with factor loadings greater than 0.30 are considered acceptable measures of the underlying factors. c) "r" represents a correlation between the Brooding and Reflection factors. d) RM1–RM10 represents questionnaire items 1–10. RM1 = Think "What am I doing to deserve this?", RM3 = "Think Why do I always react this way?", RM6 = Think about a recent situation, wishing it had gone better, RM7 = Think "Why do I have problems other people don't have?", RM8 = Think "Why can't I handle things better?", RM2 = Analyse recent events to try to understand why you are depressed, RM4 = Go away by yourself and think about why you feel this way, RM5 = Write down what you are thinking and analyse it, RM9 = Analyse your personality to try to understand why you are depressed, and RM10 = Go someplace alone to think about your feelings. e) Sample size = 747; Chi-square = 191.132; Degree of freedom = 33; $P < 0.001$; Comparative Fit Index = 0.91; RMSEA = 0.06; 90% Confidence Interval 0.07–0.09.

collected data only one time; and thus, it is inconclusive if the results are stable across time and if they can be replicable.

Results from this study have implications for nurse educators. Given that nursing students experience various stressors, they may ruminate on such stressors and then develop emotional symptoms (such as depression). Nurse educators may monitor students' mental health and may use the RRS-10 to screen their rumination levels. Students with high rumination should be referred to university counselling (or mental health professionals) for further investigation. If needed, professional help (such as counselling) can be provided to students to prevent the occurrence of mental health problems.

Nurse researchers may use the RRS-10 to assess the rumination construct in their future studies. Additional research is needed to further test the psychometric properties of the RRS-10 on Thai nursing students and other populations (such as university students from other schools, graduate students, participants from other

universities in Thailand and other countries, adults, and elderly). To strengthen external validity of research findings, multi-centred recruitment with large sample sizes, longitudinal research, and cross-cultural studies are strongly encouraged.

Conclusion

This study provided empirical evidence to support the psychometric properties (construct validity, concurrent validity, and internal consistency reliability) of the 10-item ruminative response scale on nursing undergraduate students in Thailand. The RRS-10 is considered a short, simple, and easily-administered. In addition to English-speaking populations, the scale can be used to screen rumination levels among Thai nursing students. Appropriate actions by nurse educators can then be provided to students with high levels of rumination. Future research could further test the psychometric properties of RRS-10 on

other Thai population (such as adults and elderly) or cross-validate the scale on people in other countries.

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